FinPhaser

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BIOS 8802: Final Pipeline Presentation + Demo

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Background

The Model: African Cichlids are popular models for evolution due to variation

Quirks: ZW System & Polygenic Sex Determination (+ Master Switch)

https://imperialtropicals.com/products/yellow-head-nkhata-bay-aulonocara-sp

York R, Patil C, Hulsey D, Anoruo O, Streelman T and Fernald R (2015). Evolution of bower building in Lake Malawi cichlid fish: Phylogeny, morphology, and behavior. Frontiers in Ecology and Evolution 3:18. doi: 10.3389/fevo.2015.00018

The "Black Box" of Backcrosses

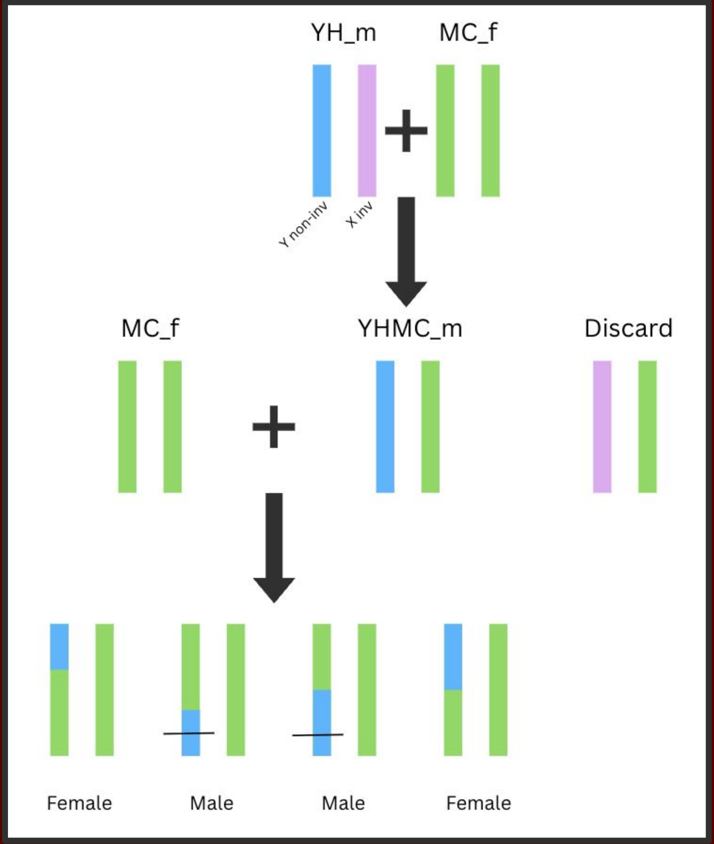
The Cross: YH x MC → BC1

The YH Male-Specific XY LG10 Inversion:

- Males are <u>heterogametic</u> on LG10
- Recombination Suppression

Verification:

- Using SPORE to check for inbreeding



"Large inversions in Lake Malawi cichlids are associated with habitat preference, lineage, and sex determination"
Kumar *et al.*, (2025)

Pipeline Flow

1

Local Ancestry Inference

Phase Parent Haplotypes

PhaseParents_VCF.py (scikit-allel)

Input VCF → Phased VCF

Reproducibility Helper

yaml_to_linkage_map.py

Prepare A_HMM Input

mod_vcf2ahmm.py

Input VCF → ancestry_input.txt
& ahmm.ploidy

Reproducibility Helper

yaml_to_popinfo.py

Run Ancestry_HMM

*ancestry_input.txt → *.posterior*
& ahmm.ploidy

2

IBD Detection

Compress VCF

bgzip, bcftools index (CSI)

Phased VCF→ comp. Phased VCF
& CSI Index

Reproducibility Helper

yaml_to_linkage_map.py

Write Sex.tsv

Reproducibility Helper

yaml_to_sex_tsv.py

samples.yml → Genomics_Sex.tsv

Run SPORE

*comp. Phased VCF→ *.ibd + etc.*
& CSI Index
& Genomics_Sex.tsv

3

Analytics

Ancestry_Summary

SNV_count.py

Input VCF → Phased VCF

Rank IBD

rank_ibd.py

**.ibd → inbreeding_rankings.txt + .tsv*

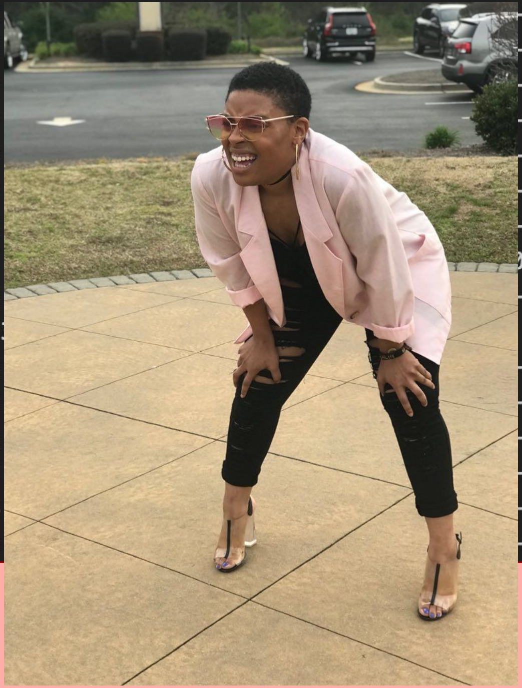
Running FinPhaser with Nextflow

- FinPhaser requires 1 input file: **samples.yml**
  - **vcf:** “data/raw/name_of_vcf.vcf”
  - **linkage_groups:** NCBI accession no. paired with LG#
  - truffle_path + SPORE parameters
  - **Populations information**
    - Parental sample id, sex, & haplotype labels
    - Offspring sample id, sex, & “admixed” label
- main.nf
  - ancestry.nf: follows the ancestry_hmm branch
  - ibd.nf: follows the spore branch
  - nextflow.config: creates `finphaser` conda env + `spore` conda env
- One command to rule them all
  - `nextflow run main.nf -profile conda`

```yaml
# —— Sample population and sex assignments ——————————————
# Parents appear TWICE (one entry per haplotype contributed).
# Offspring appear ONCE with population: admixed.
populations:

  # —— Female parent ——————————————————————
  - sample: "YH_006_f"
    population: 0        # maternal haplotype 1
    sex: F
  - sample: "YH_006_f"
    population: 1        # maternal haplotype 2
    sex: F

  # —— Male parent ———————————————————————
  - sample: "YH_011_m"
    population: 2        # paternal haplotype 1
    sex: M
  - sample: "YH_011_m"
    population: 3        # paternal haplotype 2
    sex: M

  # —— Admixed offspring ————————————————————
  - { sample: "YH_016", population: admixed, sex: F }
  - { sample: "YH_017", population: admixed, sex: F }
  - { sample: "YH_018", population: admixed, sex: F }
  - { sample: "YH_019", population: admixed, sex: F }
  - { sample: "YH_020", population: admixed, sex: F }
  - { sample: "YH_021", population: admixed, sex: F }
  - { sample: "YH_022", population: admixed, sex: F }
  - { sample: "YH_023", population: admixed, sex: F }
  - { sample: "YH_024", population: admixed, sex: F }
  - { sample: "YH_025", population: admixed, sex: M }
  - { sample: "YH_026", population: admixed, sex: M }
  - { sample: "YH_027", population: admixed, sex: M }
  - { sample: "YH_028", population: admixed, sex: M }
  - { sample: "YH_029", population: admixed, sex: M }
  - { sample: "YH_030", population: admixed, sex: M }
  - { sample: "YH_031", population: admixed, sex: F }
  - { sample: "YH_032", population: admixed, sex: F }
  - { sample: "YH_033", population: admixed, sex: F }
```

YH_016.posterior

chrom	position	2,0,0,0	1,1,0,0	1,0,1,0	1,0,0,1	0,2,0,0	0,1,1,0	0,1,0,1	0,0,2,0	0,0,1,1	0,0,0,2
LG1	100146	2.31918e-08	0.000141811	1.32928e-07	0.000662565	2.06247e-05	0.000127508	0.998681	1.2597e-08	0.000341263	2.49048e-05
LG1	101720	1.79385e-08	0.000173542	1.27734e-07	0.000523847	1.47286e-05	0.000155415	0.998849	9.40648e-09	0.000266761	1.64086e-05
LG1	104189	1.24228e-08	0.000163812	1.1763e-07	0.000516133	9.35209e-06	0.000145688	0.998895			-05
LG1	106279	9.12208e-09	0.000124451	1.0964e-07	0.000638244	6.62101e-06	0.000109955	0.998796			-06
LG1	108630	6.84788e-09	0.000118512	1.02296e-07	0.000630755	4.7773e-06	0.000104019	0.998827			-06
LG1	111345	7.05234e-09	0.000141276	1.34823e-07	0.000622032	4.71977e-06	0.000139396	0.998747			-06
LG1	113722	4.12173e-09	0.000108438	8.90147e-08	0.000614525	2.6047e-06	9.32862e-05	0.998886			-06
LG1	120566	2.13016e-09	0.000121482	7.37523e-08	0.000474187	1.02776e-06	0.00010075	0.999087			-06
LG1	123691	1.50952e-09	9.21206e-05	6.7577e-08	0.00058474	5.96712e-07	7.48706e-05	0.998988			-07
LG1	125743	1.25773e-09	0.000111844	6.4233e-08	0.000463666	4.27139e-07	8.98414e-05	0.999131			07
LG1	128514	1.01879e-09	8.58394e-05	6.00082e-08	0.000573869	2.89996e-07	6.77753e-05	0.999026			-07
LG1	130141	9.35027e-10	0.000105257	5.8108e-08	0.000457348	2.41841e-07	8.23861e-05	0.99916			
LG1	133230	8.217e-10	0.000101261	5.45866e-08	0.000454141	1.87626e-07	7.78631e-05	0.999176			-07
LG1	136038	2.62617e-09	9.75395e-05	1.22298e-07	0.00107174	1.95231e-07	7.36851e-05	0.998312			-06
LG1	144023	4.76575e-10	8.72991e-05	4.15654e-08	0.000433781	9.78483e-08	6.20579e-05	0.999249			-07
LG1	148913	3.64435e-10	6.47701e-05	3.62202e-08	0.000532054	7.60411e-08	4.3911e-05	0.999161			-07
LG1	162645	1.77463e-10	5.34004e-05	2.45889e-08	0.000504524	3.79067e-08	3.10353e-05	0.999244			-07
LG1	168167	1.33325e-10	4.99264e-05	2.10406e-08	0.000493447	2.79807e-08	2.6991e-05	0.999275			-08



SNV_Count.py → calls_YH_016.posterior.csv

```
# File: YH_016.posterior
# Summary of state calls for this individual:
#   1,0,0,1: 111514
#   0,1,1,0: 101973
#   1,0,1,0: 73632
#   0,1,0,1: 68284
#   0,0,0,2: 45214
#   0,0,2,0: 14088
#   0,0,1,1: 10316
#   2,0,0,0: 4652
#   1,1,0,0: 3012
#   0,2,0,0: 1009
#----------------------------------------
chrom,position,called_state
LG1,100146,"0,1,0,1"
LG1,101720,"0,1,0,1"
LG1,104189,"0,1,0,1"
LG1,106279,"0,1,0,1"
LG1,108630,"0,1,0,1"
LG1,111345,"0,1,0,1"
LG1,113722,"0,1,0,1"
LG1,120566,"0,1,0,1"
LG1,123691,"0,1,0,1"
LG1,125743,"0,1,0,1"
LG1,128514,"0,1,0,1"
```

```
# GLOBAL ANCESTRY SUMMARY (Aggregated over all files)
# Total SNVs called per state:
#   0,1,1,0: 2251121
#   1,0,1,0: 2105134
#   0,1,0,1: 2036380
#   1,0,0,1: 1979386
#   0,0,0,2: 742672
#   0,0,2,0: 538161
#   2,0,0,0: 259431
#   0,0,1,1: 217654
#   1,1,0,0: 158878
#   0,2,0,0: 119839
#----------------------------------------
,"0,1,0,1","0,1,1,0","1,0,0,1","1,0,1,0","0,0,0,2","0,0,2,0","0,0,1,1","1,1,0,0","2,0,0,0","0,2,0,0"
YH_024.posterior,128725,87638,67760,58256,57271,18221,8214,5660,1006,943
YH_030.posterior,76810,137734,61228,74116,32536,30340,13957,4693,1965,315
YH_029.posterior,43767,156834,67797,112726,10175,11445,11146,12064,6438,1302
YH_022.posterior,76510,163399,85162,51633,15111,23446,6551,2795,6714,2373
YH_017.posterior,102400,93532,78433,48197,40599,27045,1442,13036,22540,6470
YH_041.posterior,66712,66849,108080,118867,34780,11671,16026,6956,3653,100
YH_031.posterior,108337,76714,113669,94438,20163,7549,4298,336,8005,185
YH_025.posterior,91048,136214,62139,74315,29963,23047,3541,5294,5527,2606
YH_028.posterior,86454,55733,113846,100798,30740,19557,16024,190,9858,494
YH_037.posterior,68497,134724,65312,86614,25031,30478,346,14782,6599,1311
YH_023.posterior,99986,32763,104461,84603,47305,35762,20413,75,3113,5213
YH_016.posterior,68284,101973,111514,73632,45214,14088,10316,3012,4652,1009
YH_040.posterior,75739,90952,38450,59319,40703,36113,3941,15531,24884,48062
YH_026.posterior,65276,132244,73079,105703,17341,23004,11594,0,4772,681
YH_032.posterior,102254,115586,49880,62161,10428,43411,14912,29587,5309,166
YH_020.posterior,93234,59741,110468,106075,28933,16144,14580,2833,1681,5
YH_018.posterior,80891,85513,60284,106476,26965,28899,46,20627,20092,3901
YH_039.posterior,59637,63475,168353,98879,17409,14986,5397,0,5233,325
YH_033.posterior,105545,93831,107093,82781,23057,2953,13739,495,4069,131
YH_027.posterior,80717,95128,85711,94797,30304,29277,543,4721,7493,5003
YH_021.posterior,76443,33668,38522,89333,74410,45166,2995,3858,53343,15956
YH_019.posterior,80973,65766,103549,115056,26561,23233,12418,0,5954,184
YH_038.posterior,88956,54225,52469,95620,39239,13475,8880,12067,46020,22743
YH_042.posterior,109185,116885,52127,110739,18434,8851,16335,266,511,361
```

YHPed1 _spore.vcf.gz-truffle.ibd

ID1	ID2	NMARK	NCOMMON	IBD0	IBD1_MAX	IBD1_NSEGS	IBD1	IBD2_MAX	IBD2_NSEGS	IBD2	SEX
YH011m	YH006f	2125065	0	1.000000	2644	0	0.000000	890	0	0.000000	00
YH011m	YH019	2125065	0	1.000000	21862	0	0.000000	2683	0	0.000000	00
YH006f	YH017	2125065	0	1.000000	7530	0	0.000000	526	0	0.000000	00
YH006f	YH021	2125065	0	1.000000	5117	0	0.000000	1616	0	0.000000	00
YH016	YH020	2125065	0	0.985855	30060	1	0.014145	3007	0	0.000000	00
YH017	YH020	2125065	0	0.995926	18937	0	0.000000	4505	2	0.004074	00
YH018	YH021	2125065	0	0.986171	29387	1	0.009014	5901	2	0.004815	00
YH011m	YH022	2125065	0	1.000000	20709	0	0.000000	3593	0	0.000000	00
YH011m	YH026	2125065	0	0.988295	24874	1	0.011705	3159	0	0.000000	00
YH006f	YH022	2125065	0	1.000000	12704	0	0.000000	2001	0	0.000000	00
YH006f	YH026	2125065	0	1.000000	9936	0	0.000000	2198	0	0.000000	00
YH016	YH022	2125065	0	0.918001	35578	6	0.081999	3093	0	0.000000	00
YH016	YH026	2125065	0	0.815893	51210	10	0.181250	6071	1	0.002857	00
YH017	YH022	2125065	0	0.986786	28081	1	0.013214	2531	0	0.000000	00
YH017	YH026	2125065	0	0.872923	94468	6	0.127077	3364	0	0.000000	00
YH018	YH022	2125065	0	0.973053	31583	2	0.022554	4709	2	0.004394	00
YH018	YH026	2125065	0	0.959754	44635	2	0.037788	5222	1	0.002457	00
YH019	YH022	2125065	0	0.923014	39830	5	0.074781	4686	1	0.002205	00
YH019	YH026	2125065	0	0.842157	78920	7	0.157843	1859	0	0.000000	00
YH020	YH022	2125065	0	0.969984	34820	2	0.025552	4791	2	0.004464	00
YH020	YH026	2125065	0	0.957116	37659	3	0.042884	3228	0	0.000000	00
YH021	YH022	2125065	0	0.951551	44744	3	0.040171	6170	3	0.008278	00
YH021	YH026	2125065	0	0.949846	41063	3	0.048054	4463	1	0.002100	00
YH022	YH023	2125065	0	0.933060	48900	4	0.066940	3514	0	0.000000	00
YH022	YH027	2125065	0	0.941486	38584	4	0.049873	5331	4	0.008641	00
YH023	YH025	2125065	0	0.978554	45574	1	0.009306	7608	4	0.012140	00
YH023	YH029	2125065	0	0.987951	25604	1	0.012049	2994	0	0.000000	00
YH024	YH028	2125065	0	0.986501	28686	1	0.013499	2817	0	0.000000	00
YH025	YH028	2125065	0	0.905773	55518	5	0.094227	3531	0	0.000000	00
YH026	YH029	2125065	0	0.980970	40440	1	0.019030	3398	0	0.000000	00
YH011m	YH030	2125065	0	0.985789	30200	1	0.011849	5020	1	0.002362	00
YH011m	YH037	2125065	0	1.000000	11223	0	0.000000	2805	0	0.000000	00
YH006f	YH030	2125065	0	1.000000	6610	0	0.000000	1579	0	0.000000	00
YH006f	YH037	2125065	0	1.000000	8499	0	0.000000	2006	0	0.000000	00

inbreeding_rankings.txt

```
RANKED PAIRS BY INBREEDING POTENTIAL
====================================
Relatedness Score calculated as: (0.5 * IBD2) + (0.25 * IBD1)

  ID1    ID2     IBD0     IBD1     IBD2  Relatedness_Score Likely_Related
 YH026  YH042 0.704761 0.287468 0.007772          0.075753            YES
 YH026  YH031 0.716624 0.279033 0.004343          0.071930            YES
 YH022  YH042 0.723743 0.276257 0.000000          0.069064            YES
 YH019  YH042 0.740066 0.257562 0.002373          0.065577            YES
 YH025  YH026 0.744817 0.251794 0.003389          0.064643            YES
 YH039  YH042 0.758559 0.241441 0.000000          0.060360            YES
 YH031  YH039 0.762492 0.237508 0.000000          0.059377            YES
 YH022  YH026 0.763864 0.236136 0.000000          0.059034            YES
 YH026  YH039 0.780690 0.219310 0.000000          0.054828            YES
 YH037  YH041 0.788334 0.211666 0.000000          0.052916            YES
 YH022  YH031 0.788589 0.211411 0.000000          0.052853            YES
 YH031  YH033 0.800725 0.196294 0.002982          0.050564            YES
 YH030  YH042 0.804089 0.195911 0.000000          0.048978            YES
 YH026  YH033 0.811419 0.188581 0.000000          0.047145            YES
 YH030  YH041 0.812757 0.187243 0.000000          0.046811            YES
 YH016  YH026 0.815893 0.181250 0.002857          0.046741            YES
 YH022  YH039 0.818368 0.181632 0.000000          0.045408            YES
 YH031  YH042 0.831891 0.166092 0.002017          0.042531            YES
 YH033  YH042 0.833319 0.166681 0.000000          0.041670            YES
 YH025  YH042 0.834629 0.165371 0.000000          0.041343            YES
 YH039  YH041 0.837302 0.162698 0.000000          0.040675            YES
 YH019  YH026 0.842157 0.157843 0.000000          0.039461            YES
 YH026  YH037 0.848099 0.149529 0.002372          0.038568            YES
 YH031  YH037 0.849439 0.150561 0.000000          0.037640            YES
 YH041  YH042 0.850529 0.149471 0.000000          0.037368            YES
 YH026  YH041 0.854616 0.142574 0.002810          0.037048            YES
 YH016  YH031 0.856579 0.143421 0.000000          0.035855            YES
 YH028  YH031 0.857178 0.142822 0.000000          0.035706            YES
 YH023  YH026 0.859852 0.137658 0.002490          0.035660            YES
 YH016  YH041 0.864114 0.135886 0.000000          0.033972            YES
```